akumen | Powered by Qubiqon

APTITUDE QUESTION BANK

Computer Fundamentals

Placement preparation for IT company hiring tests

Where the Syllabus Ends and Reality Begins

150	Q1 – Q50	Q51 – Q100	Q101 – Q150
Multiple-choice questions	Beginner	Intermediate	Advanced

About this section

Computer Fundamentals covers the basics every IT professional is expected to know: hardware, memory, number systems, operating systems, networking, databases, the internet, cloud and security. These questions are usually short and factual, so they reward clear concepts more than speed. Learn the meaning of each common term and abbreviation, and understand how the parts of a computer work together.

How to use this question bank

- Part A has all 150 questions, grouped by difficulty. Attempt them on your own first, with a timer if you can (about 1 minute per question).
- Part B is the answer key. Check your answers only after finishing a full level.
- Part C explains how to solve every question in simple steps. Read the solution even for the ones you got right — it often shows a faster method.
- Mark the questions you got wrong and attempt them again after two or three days.

PART A

Questions

BEGINNER Questions 1 – 50

Q1. What does CPU stand for?

- **A)** Computer Personal Unit
- **B)** Central Program Unit
- **C)** Control Processing Utility
- **D)** Central Processing Unit

Q2. Which of these is an input device?

- **A)** Printer
- **B)** Keyboard
- **C)** Speaker
- **D)** Monitor

Q3. Which of these is an output device?

- **A)** Printer
- **B)** Mouse
- **C)** Microphone
- **D)** Scanner

Q4. 1 byte is equal to how many bits?

- **A)** 16
- **B)** 1024
- **C)** 8
- **D)** 4

Q5. What does RAM stand for?

- **A)** Random Access Memory
- **B)** Run Any Memory
- **C)** Read Access Memory
- **D)** Rapid Action Memory

Q6. Which memory loses its data when the power is switched off?

- **A)** Hard disk
- **B)** RAM
- **C)** SSD
- **D)** ROM

Q7. Which of these is an operating system?

- **A)** Photoshop
- **B)** Google Chrome
- **C)** Microsoft Word
- **D)** Windows

Q8. What is the full form of URL?

- **A)** User Resource Locator
- **B)** Universal Resource Link
- **C)** Uniform Resource Locator
- **D)** Unified Routing Location

Q9. 1 KB (kilobyte) is equal to how many bytes (in the binary system)?

- **A)** 1024
- **B)** 512
- **C)** 2048
- **D)** 1000

Q10. Which key is used to delete the character to the left of the cursor?

- **A)** Shift
- **B)** Tab
- **C)** Backspace
- **D)** Delete

Q11. Which of these is a web browser?

- **A)** Oracle
- **B)** Mozilla Firefox
- **C)** Linux
- **D)** Excel

Q12. What is the binary equivalent of the decimal number 5?

- **A)** 100
- **B)** 110
- **C)** 111
- **D)** 101

Q13. What does 'www' stand for in a website address?

- **A)** Web World Wide
- **B)** Wide World Web
- **C)** World Web Wide
- **D)** World Wide Web

Q14. Which device is used to connect a computer to the internet over telephone or cable lines?

- **A)** Modem
- **B)** Printer
- **C)** Monitor
- **D)** Keyboard

Q15. Which shortcut is commonly used to copy selected text?

- **A)** Ctrl + X
- **B)** Ctrl + C
- **C)** Ctrl + Z
- **D)** Ctrl + V

Q16. What is the main function of an operating system?

- **A)** To create documents
- **B)** To browse the internet
- **C)** To manage hardware and software resources
- **D)** To design graphics

Q17. Which of these is permanent (non-volatile) storage?

- **A)** CPU register
- **B)** RAM
- **C)** Hard Disk Drive
- **D)** Cache

Q18. What does 'email' stand for?

- **A)** Electronic mail
- **B)** Express mail
- **C)** External mail
- **D)** Easy mail

Q19. Which part of the computer performs arithmetic and logical operations?

- **A)** Hard disk
- **B)** Control Unit
- **C)** RAM
- **D)** ALU

Q20. Which file extension is used for a Microsoft Word document?

- **A)** .pptx
- **B)** .docx
- **C)** .exe
- **D)** .xlsx

Q21. What does 'PDF' stand for?

- **A)** Portable Document Format
- **B)** Personal Document File
- **C)** Printed Document File
- **D)** Public Data Format

Q22. Which of these is NOT a programming language?

- **A)** Java
- **B)** C++
- **C)** HTML
- **D)** Python

Q23. A computer virus is a type of:

- **A)** Operating system
- **B)** Malicious software
- **C)** Web browser
- **D)** Hardware

Q24. What is the brain of the computer?

- **A)** Mouse
- **B)** Monitor
- **C)** Keyboard
- **D)** CPU

Q25. Which of these is the largest unit of storage?

- **A)** Kilobyte
- **B)** Megabyte
- **C)** Gigabyte
- **D)** Terabyte

Q26. What is a 'password' used for?

- **A)** To send email
- **B)** To print documents
- **C)** To protect an account from unauthorised access
- **D)** To speed up the computer

Q27. Which device displays output in visual form?

- **A)** Speaker
- **B)** Monitor
- **C)** Keyboard
- **D)** Scanner

Q28. What does 'Wi-Fi' allow you to do?

- **A)** Connect to a network without cables
- **B)** Charge devices
- **C)** Store files permanently
- **D)** Print without paper

Q29. Which of these is an example of system software?

- **A)** Operating system
- **B)** Web browser
- **C)** Word processor
- **D)** Video game

Q30. What is the decimal value of binary 1010?

- **A)** 8
- **B)** 5
- **C)** 12
- **D)** 10

Q31. Which company developed the Android operating system (now maintained by)?

- **A)** Microsoft
- **B)** Google
- **C)** IBM
- **D)** Apple

Q32. Which storage device has no moving parts?

- **A)** HDD
- **B)** CD drive
- **C)** SSD
- **D)** Floppy drive

Q33. What is the function of the 'Ctrl + Z' shortcut?

- **A)** Close the window
- **B)** Redo
- **C)** Save the file
- **D)** Undo the last action

Q34. Which of these is a search engine?

- **A)** Windows
- **B)** Google
- **C)** Zoom
- **D)** Excel

Q35. What does 'LAN' stand for?

- **A)** Local Area Network
- **B)** Long Access Network
- **C)** Linked Area Node
- **D)** Large Area Network

Q36. Which of these is used to store data in rows and columns?

- **A)** Word processor
- **B)** Paint
- **C)** Spreadsheet
- **D)** Media player

Q37. What does 'GUI' stand for?

- **A)** Graphical User Interface
- **B)** Graphic Universal Icon
- **C)** General User Input
- **D)** Guided User Interaction

Q38. Which of these is cloud storage?

- **A)** Pen drive
- **B)** Hard disk
- **C)** Google Drive
- **D)** DVD

Q39. What is the purpose of antivirus software?

- **A)** To create presentations
- **B)** To detect and remove malicious software
- **C)** To edit photos
- **D)** To increase internet speed

Q40. Which unit is used to measure the speed of a CPU?

- **A)** Gigabyte (GB)
- **B)** Watt
- **C)** Megapixel
- **D)** Gigahertz (GHz)

Q41. What does 'USB' stand for?

- **A)** Unified System Board
- **B)** User Serial Bus
- **C)** Universal Storage Box
- **D)** Universal Serial Bus

Q42. Which of these is an example of an application software?

- **A)** Linux kernel
- **B)** MS PowerPoint
- **C)** Device driver
- **D)** BIOS

Q43. Which of these is the correct format of an email address?

- **A)** www.name@com
- **B)** name#example.com
- **C)** name@example.com
- **D)** name.example.com

Q44. What is 'booting' a computer?

- **A)** Starting up the computer and loading the OS
- **B)** Deleting files
- **C)** Installing software
- **D)** Shutting down the computer

Q45. Which of these is a valid IPv4 address?

- **A)** 192.168.1.1
- **B)** 192:168:1:1
- **C)** 192.168.1
- **D)** 256.300.1.1

Q46. What is the full form of 'HTTP'?

- **A)** Home Text Transfer Protocol
- **B)** High Transfer Text Protocol
- **C)** HyperText Transfer Protocol
- **D)** HyperText Transmission Program

Q47. What does 'Ctrl + S' do in most programs?

- **A)** Searches
- **B)** Saves the file
- **C)** Selects all
- **D)** Shuts down

Q48. Which of these is a social media platform?

- **A)** MySQL
- **B)** Unix
- **C)** Oracle
- **D)** LinkedIn

Q49. What is a 'folder' used for on a computer?

- **A)** To print files
- **B)** To run programs faster
- **C)** To organise and group files
- **D)** To connect to Wi-Fi

Q50. Which of these is used to point and click on screen items?

- **A)** Mouse
- **B)** Printer
- **C)** Speaker
- **D)** Projector

INTERMEDIATE Questions 51 – 100

Q51. What is the hexadecimal equivalent of decimal 255?

- **A)** F0
- **B)** FF
- **C)** FE
- **D)** EF

Q52. Which layer of the OSI model is responsible for routing packets between networks?

- **A)** Data link layer
- **B)** Transport layer
- **C)** Session layer
- **D)** Network layer

Q53. What is the default port number for HTTPS?

- **A)** 443
- **B)** 25
- **C)** 80
- **D)** 21

Q54. Which of these is a primary key's main property?

- **A)** It must be a text field
- **B)** It uniquely identifies each row and cannot be NULL
- **C)** It can have duplicate values
- **D)** It links two databases

Q55. What is the role of DNS?

- **A)** It speeds up the CPU
- **B)** It encrypts data
- **C)** It translates domain names into IP addresses
- **D)** It stores emails

Q56. Which memory is the fastest?

- **A)** Cache memory
- **B)** RAM
- **C)** Hard disk
- **D)** CPU registers

Q57. What does 'SQL' stand for?

- **A)** Structured Query Language
- **B)** Sequential Query Logic
- **C)** Simple Query Language
- **D)** System Query Language

Q58. Which protocol is used to send emails?

- **A)** HTTP
- **B)** SNMP
- **C)** SMTP
- **D)** FTP

Q59. What is virtual memory?

- **A)** Memory in the cloud
- **B)** Memory on the graphics card
- **C)** A type of ROM
- **D)** Using part of the hard disk as an extension of RAM

Q60. Which of these is NOT an operating system?

- **A)** Ubuntu
- **B)** MySQL
- **C)** macOS
- **D)** Windows

Q61. What is the binary of decimal 25?

- **A)** 10101
- **B)** 11001
- **C)** 11010
- **D)** 10011

Q62. What is a 'firewall'?

- **A)** A program that cools the CPU
- **B)** A backup device
- **C)** A type of virus
- **D)** A system that monitors and controls network traffic based on rules

Q63. Which of these is a NoSQL database?

- **A)** MongoDB
- **B)** MySQL
- **C)** Oracle Database
- **D)** PostgreSQL

Q64. Which device works at the Data Link layer to forward frames using MAC addresses?

- **A)** Hub
- **B)** Router
- **C)** Switch
- **D)** Repeater

Q65. What is the full form of 'IP' in IP address?

- **A)** Interface Program
- **B)** Internet Protocol
- **C)** Internal Port
- **D)** Internet Provider

Q66. Which of these is used to convert source code into machine code all at once before running?

- **A)** Interpreter
- **B)** Assembler
- **C)** Linker
- **D)** Compiler

Q67. What is phishing?

- **A)** A method of data backup
- **B)** A type of hardware failure
- **C)** Tricking people into revealing sensitive information through fake messages or websites
- **D)** A network speed test

Q68. What does 'BIOS' do?

- **A)** Performs hardware checks and starts the boot process
- **B)** Edits documents
- **C)** Stores user files
- **D)** Manages internet connection

Q69. In networking, what does 'bandwidth' refer to?

- **A)** The number of users
- **B)** The maximum data transfer rate of a connection
- **C)** The size of a file
- **D)** The length of the cable

Q70. What does 'SaaS' stand for in cloud computing?

- **A)** System as a Service
- **B)** Security as a Service
- **C)** Software as a Service
- **D)** Storage as a Service

Q71. Which SQL command is used to retrieve data from a table?

- **A)** DELETE
- **B)** UPDATE
- **C)** INSERT
- **D)** SELECT

Q72. What is the main difference between HTTP and HTTPS?

- **A)** HTTPS encrypts data using TLS/SSL
- **B)** HTTPS is faster always
- **C)** HTTPS does not use a port
- **D)** HTTP is newer

Q73. Which of these is an example of an open-source operating system?

- **A)** Windows 11
- **B)** macOS
- **C)** Linux
- **D)** iOS

Q74. What is the function of a router?

- **A)** To forward data packets between different networks
- **B)** To store files
- **C)** To print documents
- **D)** To display output

Q75. What is a 'deadlock' in an operating system?

- **A)** A crashed hard disk
- **B)** A situation where processes wait for each other's resources forever
- **C)** A locked screen
- **D)** A virus attack

Q76. What is the octal equivalent of decimal 64?

- **A)** 64
- **B)** 80
- **C)** 77
- **D)** 100

Q77. Which type of cloud service gives you virtual machines, storage and networks to manage yourself?

- **A)** DaaS
- **B)** SaaS
- **C)** PaaS
- **D)** IaaS

Q78. What does 'normalisation' in databases aim to do?

- **A)** Encrypt the database
- **B)** Make queries slower
- **C)** Reduce data redundancy and improve consistency
- **D)** Delete old data

Q79. Which of the following is a volatile memory?

- **A)** Cache
- **B)** Flash memory
- **C)** ROM
- **D)** SSD

Q80. What is two-factor authentication (2FA)?

- **A)** Logging in from two devices
- **B)** Using two different methods to verify identity, like a password and an OTP
- **C)** Using two passwords
- **D)** Having two user accounts

Q81. Which protocol automatically assigns IP addresses to devices on a network?

- **A)** FTP
- **B)** DNS
- **C)** ARP
- **D)** DHCP

Q82. What is the purpose of a 'foreign key' in a database?

- **A)** To sort data
- **B)** To make a column unique
- **C)** To link a row in one table to a row in another table
- **D)** To encrypt a table

Q83. What is the main job of the 'kernel' in an operating system?

- **A)** Browsing the internet
- **B)** Managing CPU, memory and devices at the core level
- **C)** Displaying the desktop wallpaper
- **D)** Compressing files

Q84. Which of these is a characteristic of cloud computing?

- **A)** On-demand access to resources over the internet
- **B)** No internet required
- **C)** Data stored only on your own PC
- **D)** Fixed hardware only

Q85. What does 'MAC' stand for in a MAC address?

- **A)** Machine Address Code
- **B)** Media Access Control
- **C)** Memory Access Control
- **D)** Main Access Computer

Q86. What is 'multitasking' in an operating system?

- **A)** Running multiple programs apparently at the same time
- **B)** Having many users
- **C)** Installing many programs
- **D)** Using multiple monitors

Q87. Which of these is an example of a relational database management system (RDBMS)?

- **A)** MongoDB
- **B)** Redis
- **C)** Cassandra
- **D)** MySQL

Q88. How many bits are in an IPv4 address?

- **A)** 128
- **B)** 64
- **C)** 32
- **D)** 16

Q89. What is 'encryption'?

- **A)** Copying data
- **B)** Converting data into a coded form so only authorised people can read it
- **C)** Compressing data
- **D)** Deleting data permanently

Q90. What does the 'Control Unit' of the CPU do?

- **A)** Displays output
- **B)** Stores files
- **C)** Directs the operation of the processor and coordinates components
- **D)** Performs only addition

Q91. Which topology connects all devices to a single central hub or switch?

- **A)** Star
- **B)** Mesh
- **C)** Bus
- **D)** Ring

Q92. What is a 'cookie' in web browsing?

- **A)** A type of image
- **B)** A virus
- **C)** A browser plug-in
- **D)** A small file a website stores in your browser to remember information

Q93. Which generation of computers used integrated circuits (ICs)?

- **A)** Fifth generation
- **B)** First generation
- **C)** Second generation
- **D)** Third generation

Q94. What does 'API' stand for?

- **A)** Automated Programming Input
- **B)** Application Programming Interface
- **C)** Application Process Integration
- **D)** Advanced Program Instruction

Q95. Which command in Windows shows the IP configuration of the computer?

- **A)** ping
- **B)** netstat
- **C)** ipconfig
- **D)** tracert

Q96. What is the main advantage of an SSD over an HDD?

- **A)** Much faster read/write speeds
- **B)** Lower cost per GB always
- **C)** Larger physical size
- **D)** Uses magnetic platters

Q97. What is 'ransomware'?

- **A)** Free antivirus software
- **B)** Malware that locks or encrypts your files and demands payment
- **C)** A cloud storage service
- **D)** A backup tool

Q98. What is the result of the Boolean expression TRUE AND FALSE?

- **A)** NULL
- **B)** TRUE
- **C)** Error
- **D)** FALSE

Q99. Which of these is a version control system?

A) Nginx
B) Jenkins
C) Git
D) Docker

Q100. In which unit is the resolution of a screen usually described?

A) Pixels
B) Bytes
C) Hertz
D) Watts

ADVANCED Questions 101 – 150

Q101. Which page-replacement algorithm can suffer from Belady's anomaly (more frames causing more page faults)?

- **A)** Optimal
- **B)** FIFO
- **C)** LFU
- **D)** LRU

Q102. Which normal form removes transitive dependencies?

- **A)** Third Normal Form (3NF)
- **B)** First Normal Form
- **C)** Fourth Normal Form
- **D)** Second Normal Form

Q103. In the TCP/IP model, which protocol provides reliable, connection-oriented delivery?

- **A)** IP
- **B)** ICMP
- **C)** TCP
- **D)** UDP

Q104. What is the subnet mask for a /24 network?

- **A)** 255.255.0.0
- **B)** 255.0.0.0
- **C)** 255.255.255.255
- **D)** 255.255.255.0

Q105. How many usable host addresses are there in a /26 IPv4 subnet?

- **A)** 64
- **B)** 62
- **C)** 126
- **D)** 30

Q106. Which of the ACID properties ensures that a transaction is all-or-nothing?

- **A)** Isolation
- **B)** Durability
- **C)** Consistency
- **D)** Atomicity

Q107. Which scheduling algorithm can cause starvation of long processes?

- **A)** Shortest Job First
- **B)** First Come First Served
- **C)** Round Robin
- **D)** Multilevel with ageing

Q108. What is the 2's complement representation of −5 in 8 bits?

- **A)** 10000101
- **B)** 11111010
- **C)** 11111011
- **D)** 00000101

Q109. Which SQL clause is used to filter groups after aggregation?

- **A)** HAVING
- **B)** GROUP BY
- **C)** WHERE
- **D)** ORDER BY

Q110. What is 'thrashing' in an operating system?

- **A)** A virus spreading
- **B)** Overheating of the CPU
- **C)** A hard disk crash
- **D)** Excessive paging that leaves the CPU little time for real work

Q111. Which cloud deployment model combines on-premises infrastructure with public cloud?

- **A)** Private cloud
- **B)** Hybrid cloud
- **C)** Community cloud
- **D)** Public cloud

Q112. Which encryption type uses a pair of keys — public and private?

- **A)** Hashing
- **B)** Symmetric encryption
- **C)** Asymmetric encryption
- **D)** Encoding

Q113. What is the main purpose of a hash function like SHA-256 in security?

- **A)** To produce a fixed-size fingerprint of data that changes if the data changes
- **B)** To encrypt and later decrypt data
- **C)** To speed up networks
- **D)** To compress files for storage

Q114. Which join returns all rows from the left table and matching rows from the right table?

- **A)** INNER JOIN
- **B)** CROSS JOIN
- **C)** RIGHT JOIN
- **D)** LEFT JOIN

Q115. What is the time complexity of searching in a balanced binary search tree?

- **A)** O(n)
- **B)** O(log n)
- **C)** O(n log n)
- **D)** O(1)

Q116. Which OSI layer handles encryption, compression and data format translation?

- **A)** Transport layer
- **B)** Session layer
- **C)** Presentation layer
- **D)** Application layer

Q117. What does a 'semaphore' help with in an operating system?

- **A)** Controlling access to shared resources by multiple processes
- **B)** Connecting to Wi-Fi
- **C)** Displaying graphics
- **D)** Speeding up the hard disk

Q118. Which of the following is an example of a SQL injection attack input?

- **A)** Hello
- **B)** ping 8.8.8.8
- **C)** ' OR '1'='1
- **D)** SELECT * FROM users

Q119. In virtualisation, what does a hypervisor do?

- **A)** Manages email traffic
- **B)** Creates and runs virtual machines on physical hardware
- **C)** Encrypts the hard disk
- **D)** Acts as a web server

Q120. What is the key difference between a process and a thread?

- **A)** Processes are faster than threads
- **B)** There is no difference
- **C)** Threads cannot run at the same time
- **D)** Threads of the same process share memory; processes have separate memory

Q121. Which RAID level uses mirroring for redundancy?

- **A)** RAID 0
- **B)** RAID 1
- **C)** RAID 10 only
- **D)** RAID 5

Q122. What is the purpose of the 'ARP' protocol?

- **A)** To find the MAC address for a given IP address
- **B)** To assign IP addresses
- **C)** To resolve domain names
- **D)** To transfer files

Q123. What is a 'race condition'?

- **A)** A sorting algorithm
- **B)** A very fast CPU
- **C)** A bug where the result depends on the timing of concurrent operations
- **D)** A network speed test

Q124. Which of these is a container orchestration platform?

- **A)** Apache Kafka
- **B)** MySQL
- **C)** Git
- **D)** Kubernetes

Q125. Which SQL statement removes all rows from a table quickly but keeps its structure?

- **A)** DROP
- **B)** REMOVE
- **C)** DELETE with WHERE
- **D)** TRUNCATE

Q126. What does CIA stand for in information security?

- **A)** Confidentiality, Integrity, Availability
- **B)** Code, Information, Audit
- **C)** Control, Identity, Access
- **D)** Cyber, Internet, Authentication

Q127. What is the hexadecimal value of binary 11011110?

- **A)** DF
- **B)** DE
- **C)** ED
- **D)** CE

Q128. Which memory management technique divides memory into fixed-size blocks called frames?

- **A)** Fragmentation
- **B)** Swapping
- **C)** Paging
- **D)** Segmentation

Q129. What is 'latency' in networking?

- **A)** The cable length
- **B)** The number of connected devices
- **C)** The time delay for data to travel from source to destination
- **D)** The total data transferred

Q130. Which of these is an example of PaaS?

- **A)** Google App Engine
- **B)** Dropbox
- **C)** Gmail
- **D)** Amazon EC2

Q131. In a database, what is an 'index' used for?

- **A)** To encrypt data
- **B)** To speed up data retrieval
- **C)** To delete duplicates automatically
- **D)** To store backups

Q132. What is the default port number of SSH?

- **A)** 23
- **B)** 3389
- **C)** 21
- **D)** 22

Q133. Which of these best describes 'zero trust' security?

- **A)** Disable all passwords
- **B)** Never trust by default; always verify every user and device
- **C)** Use only one firewall
- **D)** Trust everyone inside the network

Q134. What does a CDN (Content Delivery Network) do?

- **A)** Serves content from servers close to the user to reduce delay
- **B)** Manages databases
- **C)** Compiles code
- **D)** Stores passwords

Q135. Which component of a computer translates virtual addresses into physical addresses?

- **A)** ALU
- **B)** DMA controller
- **C)** MMU (Memory Management Unit)
- **D)** GPU

Q136. In SQL, which of these correctly counts rows in a table named 'employees'?

- **A)** SELECT SUM(*) FROM employees;
- **B)** SELECT TOTAL FROM employees;
- **C)** COUNT ROWS employees;
- **D)** SELECT COUNT(*) FROM employees;

Q137. What is the main benefit of using a load balancer?

- **A)** Distributing traffic across servers to improve availability and performance
- **B)** Encrypting data
- **C)** Replacing the database
- **D)** Storing files

Q138. Which protocol is connectionless and used for fast, real-time traffic like video calls?

- **A)** FTP
- **B)** SMTP
- **C)** TCP
- **D)** UDP

Q139. What does 'DMA' allow?

- **A)** Disk encryption
- **B)** Devices to transfer data to memory without constant CPU involvement
- **C)** Direct internet access
- **D)** Faster typing

Q140. What is the result of the bitwise operation 12 & 10?

- **A)** 6
- **B)** 2
- **C)** 8
- **D)** 14

Q141. What is the purpose of a VPN?

- **A)** To block all websites
- **B)** To store files offline
- **C)** To create an encrypted tunnel over a public network
- **D)** To speed up the CPU

Q142. Which of these is a characteristic of a microservices architecture?

- **A)** The application is built as small, independently deployable services
- **B)** It cannot use APIs
- **C)** All code is in one large program
- **D)** It runs only on one server

Q143. What is 'paging' different from 'segmentation' in?

- **A)** Paging is only for disks
- **B)** Paging uses fixed-size blocks; segmentation uses variable-size logical units
- **C)** They are the same
- **D)** Segmentation is faster always

Q144. Which of these is an example of a public cloud provider?

- **A)** A home NAS
- **B)** A company's in-house server room
- **C)** A personal laptop
- **D)** Microsoft Azure

Q145. What does the SQL keyword DISTINCT do?

- **A)** Returns only unique values
- **B)** Counts rows
- **C)** Deletes duplicates from the table
- **D)** Sorts the result

Q146. Which of these is the IPv6 loopback address?

- **A)** FF::FF
- **B)** ::1
- **C)** 0.0.0.0
- **D)** 127.0.0.1

Q147. What is 'DevOps' primarily about?

- **A)** A programming language
- **B)** Designing hardware
- **C)** Collaboration between development and operations to deliver software faster and more reliably
- **D)** Only writing documentation

Q148. What is the output of the logical expression NOT (TRUE OR FALSE)?

- **A)** Error
- **B)** NULL
- **C)** TRUE
- **D)** FALSE

Q149. What does 'idempotent' mean for an HTTP method like PUT?

- **A)** It creates a new resource every time
- **B)** It can never fail
- **C)** Making the same request many times has the same effect as making it once
- **D)** It is always encrypted

Q150. Which algorithm is commonly used to securely exchange keys over a public channel?

- **A)** Diffie–Hellman
- **B)** Dijkstra
- **C)** Huffman coding
- **D)** Bubble sort

PART B

Answer Key

BEGINNER Questions 1 – 50

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	D	2	B	3	A	4	C	5	A
6	B	7	D	8	C	9	A	10	C
11	B	12	D	13	D	14	A	15	B
16	C	17	C	18	A	19	D	20	B
21	A	22	C	23	B	24	D	25	D
26	C	27	B	28	A	29	A	30	D
31	B	32	C	33	D	34	B	35	A
36	C	37	A	38	C	39	B	40	D
41	D	42	B	43	C	44	A	45	A
46	C	47	B	48	D	49	C	50	A

INTERMEDIATE Questions 51 – 100

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
51	B	52	D	53	A	54	B	55	C
56	D	57	A	58	C	59	D	60	B
61	B	62	D	63	A	64	C	65	B
66	D	67	C	68	A	69	B	70	C
71	D	72	A	73	C	74	A	75	B
76	D	77	D	78	C	79	A	80	B
81	D	82	C	83	B	84	A	85	B
86	A	87	D	88	C	89	B	90	C
91	A	92	D	93	D	94	B	95	C
96	A	97	B	98	D	99	C	100	A

ADVANCED Questions 101 – 150

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
101	B	102	A	103	C	104	D	105	B
106	D	107	A	108	C	109	A	110	D
111	B	112	C	113	A	114	D	115	B
116	C	117	A	118	C	119	B	120	D
121	B	122	A	123	C	124	D	125	D
126	A	127	B	128	C	129	C	130	A

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
131	B	132	D	133	B	134	A	135	C
136	D	137	A	138	D	139	B	140	C
141	C	142	A	143	B	144	D	145	A
146	B	147	C	148	D	149	C	150	A

PART C

How to Solve — Step-by-Step Solutions

Each solution shows the correct option and a simple way to reach it. Try to understand the method, not just the answer — the same method works for many similar questions.

BEGINNER Questions 1 – 50

Q1 Answer: D) Central Processing Unit

CPU = Central Processing Unit, the 'brain' that carries out instructions.

Q2 Answer: B) Keyboard

An input device sends data into the computer. A keyboard does this; the others show or produce output.

Q3 Answer: A) Printer

A printer produces output (a printed page). The others are input devices.

Q4 Answer: C) 8

1 byte = 8 bits. A bit is a single 0 or 1.

Q5 Answer: A) Random Access Memory

RAM = Random Access Memory, the temporary working memory of a computer.

Q6 Answer: B) RAM

RAM is volatile — its contents are lost without power. ROM, HDD and SSD keep data.

Q7 Answer: D) Windows

Windows manages the hardware and runs other programs. Word, Chrome and Photoshop are applications.

Q8 Answer: C) Uniform Resource Locator

URL = Uniform Resource Locator — the address of a web page, e.g. https://www.akumenbyq.com.

Q9 Answer: A) 1024

In computing, 1 KB = 2^{10} = 1024 bytes.

Q10 Answer: C) Backspace

Backspace removes the character to the left; Delete removes the one to the right.

Q11 Answer: B) Mozilla Firefox

Firefox is a browser used to view websites. Linux is an OS, Excel a spreadsheet, Oracle a database.

Q12 Answer: D) 101

5 = 4 + 1 = $1\times2^2 + 0\times2^1 + 1\times2^0$ → 101.

Q13 Answer: D) World Wide Web

www = World Wide Web.

Q14 **Answer: A) Modem**

A modem (modulator–demodulator) converts digital signals so they can travel over lines to the internet.

Q15 **Answer: B) Ctrl + C**

Ctrl + C copies, Ctrl + V pastes, Ctrl + X cuts, Ctrl + Z undoes.

Q16 **Answer: C) To manage hardware and software resources**

The OS manages memory, files, devices and programs so the computer can be used.

Q17 **Answer: C) Hard Disk Drive**

A hard disk keeps data even when the computer is off.

Q18 **Answer: A) Electronic mail**

Email = electronic mail, messages sent over a network.

Q19 **Answer: D) ALU**

The ALU (Arithmetic Logic Unit) inside the CPU does calculations and comparisons.

Q20 **Answer: B) .docx**

.docx is Word, .xlsx is Excel, .pptx is PowerPoint, .exe is a program file.

Q21 **Answer: A) Portable Document Format**

PDF = Portable Document Format, which looks the same on any device.

Q22 **Answer: C) HTML**

HTML is a markup language that structures web pages; it has no programming logic like loops or conditions.

Q23 **Answer: B) Malicious software**

A virus is malware — software designed to harm or misuse a computer.

Q24 **Answer: D) CPU**

The CPU processes all instructions, so it is called the brain of the computer.

Q25 **Answer: D) Terabyte**

Order: KB < MB < GB < TB. Terabyte is the largest here.

Q26 **Answer: C) To protect an account from unauthorised access**

A password verifies your identity so others cannot access your account.

Q27 **Answer: B) Monitor**

A monitor shows text, images and video.

Q28 **Answer: A) Connect to a network without cables**

Wi-Fi is wireless networking — connecting to the internet or a local network without cables.

Q29 **Answer: A) Operating system**

System software (like the OS) runs the computer. The others are application software.

Q30 **Answer: D) 10**

1010 = 1×8 + 0×4 + 1×2 + 0×1 = 10.

Q31 **Answer: B) Google**

Android is developed and maintained by Google.

Q32 **Answer: C) SSD**

A Solid State Drive uses flash memory chips with no spinning parts, so it is faster and more durable.

Q33 **Answer: D) Undo the last action**

Ctrl + Z undoes the most recent action.

Q34 **Answer: B) Google**

Google helps you search the web. Windows is an OS, Excel a spreadsheet and Zoom a video app.

Q35 **Answer: A) Local Area Network**

LAN = Local Area Network, connecting computers in a small area like an office.

Q36 **Answer: C) Spreadsheet**

Spreadsheets (like Excel) organise data in rows and columns.

Q37 **Answer: A) Graphical User Interface**

GUI = Graphical User Interface — windows, icons and menus instead of typed commands.

Q38 **Answer: C) Google Drive**

Google Drive stores files on internet servers (the cloud).

Q39 **Answer: B) To detect and remove malicious software**

Antivirus software finds and removes viruses and other malware.

Q40 **Answer: D) Gigahertz (GHz)**

CPU clock speed is measured in hertz, usually GHz (billions of cycles per second).

Q41 **Answer: D) Universal Serial Bus**

USB = Universal Serial Bus, a standard port to connect devices.

Q42 **Answer: B) MS PowerPoint**

PowerPoint is used by people to do a task (presentations). The others are system-level software.

Q43 **Answer: C) name@example.com**

An email address has a user name, the '@' symbol and a domain: name@example.com.

Q44 **Answer: A) Starting up the computer and loading the OS**

Booting is the start-up process in which the OS is loaded into memory.

Q45 **Answer: A) 192.168.1.1**

IPv4 has four numbers (0–255) separated by dots. 192.168.1.1 is valid.

Q46 **Answer: C) HyperText Transfer Protocol**

HTTP = HyperText Transfer Protocol, used to transfer web pages.

Q47 **Answer: B) Saves the file**

Ctrl + S saves the current file.

Q48 **Answer: D) LinkedIn**

LinkedIn is a professional social network. The others are database or OS software.

Q49 **Answer: C) To organise and group files**

Folders (directories) hold files and other folders to keep things organised.

Q50 **Answer: A) Mouse**

A mouse moves the pointer and selects items by clicking.

INTERMEDIATE Questions 51 – 100

Q51 Answer: B) FF

255 = 15 × 16 + 15. In hexadecimal, 15 is F → FF.

Q52 Answer: D) Network layer

The Network layer (Layer 3) handles logical addressing (IP) and routing.

Q53 Answer: A) 443

HTTPS uses port 443. HTTP uses 80, FTP 21, SMTP 25.

Q54 Answer: B) It uniquely identifies each row and cannot be NULL

A primary key uniquely identifies every record in a table and must not be empty (NULL).

Q55 Answer: C) It translates domain names into IP addresses

DNS (Domain Name System) converts names like google.com into IP addresses that computers use.

Q56 Answer: D) CPU registers

Speed order: registers > cache > RAM > hard disk. Registers are inside the CPU itself.

Q57 Answer: A) Structured Query Language

SQL = Structured Query Language, used to manage data in relational databases.

Q58 Answer: C) SMTP

SMTP (Simple Mail Transfer Protocol) sends emails. POP3/IMAP are used to receive them.

Q59 Answer: D) Using part of the hard disk as an extension of RAM

When RAM is full, the OS moves less-used data to the disk (page/swap file), making memory appear larger.

Q60 Answer: B) MySQL

MySQL is database software. Ubuntu, macOS and Windows are operating systems.

Q61 Answer: B) 11001

25 = 16 + 8 + 1 → 1×16 + 1×8 + 0×4 + 0×2 + 1×1 = 11001.

Q62 Answer: D) A system that monitors and controls network traffic based on rules

A firewall allows or blocks network traffic to protect a system from unauthorised access.

Q63 Answer: A) MongoDB

MongoDB stores data as documents (JSON-like) and is a NoSQL database. The others are relational.

Q64 Answer: C) Switch

A switch uses MAC addresses to send frames only to the correct port. Routers use IP addresses (Layer 3).

Q65 Answer: B) Internet Protocol

IP = Internet Protocol.

Q66 Answer: D) Compiler

A compiler translates the whole program before execution. An interpreter translates and runs line by line.

Q67 Answer: C) Tricking people into revealing sensitive information through fake messages or websites

Phishing uses fake emails, messages or sites that look genuine to steal passwords or money.

Q68 Answer: A) Performs hardware checks and starts the boot process

BIOS (Basic Input/Output System) runs on start-up, tests hardware (POST) and loads the bootloader.

Q69 Answer: B) The maximum data transfer rate of a connection

Bandwidth is how much data can be sent per second, e.g. 100 Mbps.

Q70 Answer: C) Software as a Service

SaaS delivers ready-to-use software over the internet (e.g. Gmail, Microsoft 365).

Q71 Answer: D) SELECT

SELECT reads data. INSERT adds, UPDATE changes and DELETE removes rows.

Q72 Answer: A) HTTPS encrypts data using TLS/SSL

The 'S' means secure: HTTPS encrypts traffic between browser and server.

Q73 Answer: C) Linux

Linux's source code is freely available to view and modify.

Q74 Answer: A) To forward data packets between different networks

A router connects networks (e.g. your home network to the internet) and chooses the path for data.

Q75 Answer: B) A situation where processes wait for each other's resources forever

Deadlock happens when each process holds a resource and waits for one held by another, so none can continue.

Q76 Answer: D) 100

$64 = 1 \times 8^2 + 0 \times 8 + 0 \rightarrow 100$ in octal.

Q77 Answer: D) IaaS

IaaS (Infrastructure as a Service) provides raw computing resources like VMs (e.g. AWS EC2, Azure VMs).

Q78 Answer: C) Reduce data redundancy and improve consistency

Normalisation organises tables to avoid repeated data and update problems.

Q79 Answer: A) Cache

Cache memory, like RAM, loses data when power is off.

Q80 Answer: B) Using two different methods to verify identity, like a password and an OTP

2FA combines something you know (password) with something you have (phone/OTP) or are (fingerprint).

Q81 **Answer: D) DHCP**

DHCP (Dynamic Host Configuration Protocol) gives devices an IP address automatically.

Q82 **Answer: C) To link a row in one table to a row in another table**

A foreign key refers to the primary key of another table, creating a relationship.

Q83 **Answer: B) Managing CPU, memory and devices at the core level**

The kernel is the core of the OS that controls hardware resources and processes.

Q84 **Answer: A) On-demand access to resources over the internet**

Cloud services can be scaled up or down on demand and accessed over the internet.

Q85 **Answer: B) Media Access Control**

MAC = Media Access Control, a hardware address built into a network card.

Q86 **Answer: A) Running multiple programs apparently at the same time**

The OS switches the CPU quickly between tasks so several programs seem to run at once.

Q87 **Answer: D) MySQL**

MySQL stores data in related tables using SQL. The others are NoSQL systems.

Q88 **Answer: C) 32**

IPv4 addresses have 32 bits (4 bytes). IPv6 addresses have 128 bits.

Q89 **Answer: B) Converting data into a coded form so only authorised people can read it**

Encryption scrambles data using a key so it is unreadable without the matching key.

Q90 **Answer: C) Directs the operation of the processor and coordinates components**

The Control Unit fetches and decodes instructions and tells other parts what to do.

Q91 **Answer: A) Star**

In a star topology every device connects to a central device.

Q92 **Answer: D) A small file a website stores in your browser to remember information**

Cookies store small pieces of data like login sessions or preferences.

Q93 **Answer: D) Third generation**

1st: vacuum tubes, 2nd: transistors, 3rd: integrated circuits, 4th: microprocessors.

Q94 **Answer: B) Application Programming Interface**

An API defines how software components talk to each other.

Q95 **Answer: C) ipconfig**

ipconfig shows IP address, subnet mask and gateway. ping tests reachability.

Q96 **Answer: A) Much faster read/write speeds**

SSDs use flash memory, so they are much faster and quieter than spinning HDDs.

Q97 **Answer: B) Malware that locks or encrypts your files and demands payment**

Ransomware encrypts data and asks for money (ransom) to unlock it.

Q98 **Answer: D) FALSE**

AND is true only when both inputs are true. TRUE AND FALSE = FALSE.

Q99 **Answer: C) Git**

Git tracks changes to code over time. Docker is containers, Jenkins is CI/CD, Nginx is a web server.

Q100 **Answer: A) Pixels**

Screen resolution is width × height in pixels, e.g. 1920 × 1080.

ADVANCED Questions 101 – 150

Q101 **Answer: B) FIFO**

FIFO can show Belady's anomaly. Stack algorithms such as LRU and Optimal never do.

Q102 **Answer: A) Third Normal Form (3NF)**

3NF requires that non-key columns depend only on the key, not on other non-key columns (no transitive dependency).

Q103 **Answer: C) TCP**

TCP sets up a connection, numbers packets, acknowledges them and resends lost ones. UDP does not.

Q104 **Answer: D) 255.255.255.0**

/24 means the first 24 bits are 1s: 11111111.11111111.11111111.00000000 = 255.255.255.0.

Q105 **Answer: B) 62**

/26 leaves 32 − 26 = 6 host bits → 2^6 = 64 addresses. Subtract network and broadcast: 62 usable.

Q106 **Answer: D) Atomicity**

Atomicity means either every step of the transaction happens or none does.

Q107 **Answer: A) Shortest Job First**

SJF always picks the shortest job, so long jobs may wait indefinitely if short jobs keep arriving.

Q108 **Answer: C) 11111011**

+5 = 00000101. Invert: 11111010. Add 1: 11111011.

Q109 **Answer: A) HAVING**

WHERE filters rows before grouping; HAVING filters groups after GROUP BY (e.g. HAVING COUNT(*) > 5).

Q110 **Answer: D) Excessive paging that leaves the CPU little time for real work**

When too little memory is available, the system spends most of its time swapping pages in and out.

Q111 **Answer: B) Hybrid cloud**

A hybrid cloud mixes private (on-premises) and public cloud resources.

Q112 **Answer: C) Asymmetric encryption**

Asymmetric (public-key) encryption, e.g. RSA, encrypts with one key and decrypts with the other.

Q113 **Answer: A) To produce a fixed-size fingerprint of data that changes if the data changes**

Hashes are one-way. They are used to check integrity and store passwords safely; they are not meant to be decrypted.

Q114 **Answer: D) LEFT JOIN**

LEFT (OUTER) JOIN keeps every left row; where there is no match, the right-side columns are NULL.

Q115 **Answer: B) O(log n)**

A balanced BST halves the search space at each level, giving O(log n).

Q116 **Answer: C) Presentation layer**

The Presentation layer (Layer 6) deals with data representation, encryption and compression.

Q117 **Answer: A) Controlling access to shared resources by multiple processes**

A semaphore is a counter used to synchronise processes and avoid conflicts over shared resources.

Q118 **Answer: C) ' OR '1'='1**

Inserting ' OR '1'='1 into a login field can make a badly written query always true and bypass the password check.

Q119 **Answer: B) Creates and runs virtual machines on physical hardware**

A hypervisor (e.g. VMware ESXi, Hyper-V) shares physical hardware among multiple virtual machines.

Q120 **Answer: D) Threads of the same process share memory; processes have separate memory**

Threads are lighter units inside a process and share its address space; separate processes are isolated.

Q121 **Answer: B) RAID 1**

RAID 1 writes the same data on two disks (mirroring). RAID 0 is striping without redundancy.

Q122 **Answer: A) To find the MAC address for a given IP address**

ARP (Address Resolution Protocol) maps IP addresses to MAC addresses on a local network.

Q123 **Answer: C) A bug where the result depends on the timing of concurrent operations**

When two threads update shared data at the same time without proper locking, the result may be unpredictable.

Q124 **Answer: D) Kubernetes**

Kubernetes automates deployment, scaling and management of containers.

Q125 **Answer: D) TRUNCATE**

TRUNCATE empties the table quickly; DROP deletes the table itself.

Q126 **Answer: A) Confidentiality, Integrity, Availability**

The CIA triad: keep data private (confidentiality), accurate (integrity) and accessible (availability).

Q127 **Answer: B) DE**

Split into nibbles: 1101 = D, 1110 = E → DE.

Q128 **Answer: C) Paging**

Paging splits physical memory into frames and processes into pages of the same size.

Q129 **Answer: C) The time delay for data to travel from source to destination**

Latency is delay, usually measured in milliseconds (ms).

Q130 **Answer: A) Google App Engine**

PaaS provides a platform to build and run apps without managing servers, e.g. Google App Engine, Azure App Service.

Q131 **Answer: B) To speed up data retrieval**

An index works like a book's index — it helps the database find rows faster, at the cost of extra storage and slower writes.

Q132 **Answer: D) 22**

SSH uses port 22. Telnet uses 23, FTP 21, RDP 3389.

Q133 **Answer: B) Never trust by default; always verify every user and device**

Zero trust assumes no implicit trust, even inside the network; every access request is verified.

Q134 **Answer: A) Serves content from servers close to the user to reduce delay**

A CDN copies content to many locations so users get it from a nearby server, faster.

Q135 **Answer: C) MMU (Memory Management Unit)**

The MMU converts virtual addresses used by programs into real physical memory addresses.

Q136 **Answer: D) SELECT COUNT(*) FROM employees;**

COUNT(*) returns the number of rows.

Q137 **Answer: A) Distributing traffic across servers to improve availability and performance**

Load balancers spread requests so no single server is overloaded and failures are handled.

Q138 **Answer: D) UDP**

UDP sends packets without setting up a connection or retransmitting, which suits real-time media.

Q139 **Answer: B) Devices to transfer data to memory without constant CPU involvement**

Direct Memory Access lets devices like disks move data to RAM while the CPU does other work.

Q140 **Answer: C) 8**

12 = 1100 and 10 = 1010. AND bit by bit: 1000 = 8.

Q141 **Answer: C) To create an encrypted tunnel over a public network**

A VPN encrypts your traffic and routes it through a secure server, protecting it on public networks.

Q142 **Answer: A) The application is built as small, independently deployable services**

Microservices split an app into small services that communicate via APIs and can be deployed separately.

Q143 **Answer: B) Paging uses fixed-size blocks; segmentation uses variable-size logical units**

Pages are equal-sized; segments match logical parts of a program (code, stack) and vary in size.

Q144 **Answer: D) Microsoft Azure**

Azure (like AWS and Google Cloud) offers cloud services to the public over the internet.

Q145 **Answer: A) Returns only unique values**

SELECT DISTINCT removes duplicate values from the query result (not from the table).

Q146 **Answer: B) ::1**

In IPv6 the loopback address is ::1. In IPv4 it is 127.0.0.1.

Q147 **Answer: C) Collaboration between development and operations to deliver software faster and more reliably**

DevOps combines culture and tools (CI/CD, automation, monitoring) to shorten delivery cycles.

Q148 **Answer: D) FALSE**

TRUE OR FALSE = TRUE. NOT TRUE = FALSE.

Q149 **Answer: C) Making the same request many times has the same effect as making it once**

PUT sets a resource to a given state; repeating it does not change the result further.

Q150 **Answer: A) Diffie–Hellman**

Diffie–Hellman lets two parties create a shared secret key without sending it directly.